A canonical symmetry excludes the ramified quartic completion

Abstract

An explicit all-positive valid source has no matrix selector polynomial in the square root of the Bob half-angle coordinate. Four positive-definite integer block Hankel matrices give an exact separating inequality. A physical exchange of Alice and Bob transports a source inside the ramified completion cone to this excluded source. The same transformation moves Bob quarter-power testing functions to Alice and produces two incomparable certificate cones.

1 The counterexample and its validity

Use the source order $c = (d, \lambda, n, m, C, A, D, \alpha, \beta) \in \mathbb{R}^9_+$ and the complete canonical parent F_c. The cone $\mathcal{C}_{\mathrm{can}}$ consists of sources valid at every state and every pair of projective measurement angles. The two-completion matrices are

$$G_J = \begin{pmatrix} P+K & J \\ J^T & N+K \end{pmatrix}, \qquad H_J = \begin{pmatrix} P+E & K-E-J \\ K-E-J^T & N+E \end{pmatrix}.$$

Here P, N, E, K are exactly the source-affine quadratic matrices in the canonical completion note [1]. The fixed cone $\mathcal{C}_{\mathrm{ram}}$ requires $v = 16s^2$, $s \geq 0$, and one matrix polynomial $J(s) = \sum_{k=0}^{4} J_k s^k$ making both matrices positive semidefinite. The resulting Gram pencil has order 49.

The explicit Gram matrices in Ref. [1] give

$$\bar{c} = \left(1, 2^{-34}, 2^{-14}, 2^{-24}, \frac{1}{576}, 161, \frac{1}{2}, 76, \frac{1}{250}\right) \in \mathcal{C}_{\mathrm{ram}} \subseteq \mathcal{C}_{\mathrm{can}}. \tag{1}$$

Theorem 1. *The all-positive rational source*

$$\boxed{c^\dagger = \left(1, \frac{1}{576}, 161, 2^{-24}, 2^{-34}, 2^{-14}, \frac{1}{2}, \frac{1}{250}, 76\right)} \tag{2}$$

belongs to $\mathcal{C}_{\mathrm{can}} \setminus \mathcal{C}_{\mathrm{ram}}$. Consequently $\mathcal{C}_{\mathrm{ram}} \subsetneq \mathcal{C}_{\mathrm{can}}$. The source $c^\dagger$ admits no selector polynomial in $s = \sqrt{v}/4$, of any degree. The order-392 eight-copy outer lift is strictly larger than Q_{222} by the polar assembly theorem [2].

Define the involution

$$\pi c = (d, C, A, m, \lambda, n, D, \beta, \alpha), \qquad S(W, X, Y, Z) = (W, Y, X, Z).$$

The full parent satisfies the exact identity

$$\boxed{u^2 v^2 F_{\pi c}(-1/v, -1/u; S\psi) = F_c(u, v; \psi).} \tag{3}$$

The eight event packets, in the source order $(\lambda, n, m, C, A, D, \alpha, \beta)$, are carried respectively to $(C, A, m, \lambda, n, D, \beta, \alpha)$. The m, D roots change sign. The d coefficient is unchanged: its three positive packets are permuted or fixed, and $D_u D_v Z^2$ is unchanged after clearing the angles. Thus (3) holds for every source.

With homogeneous angle pairs $a = (a_0, a_1)$, $b = (b_0, b_1)$, identity (3) is the polynomial identity

$$\boxed{F^h_{\pi c}\big((b_1, -b_0), (a_1, -a_0); S\psi\big) = F^h_c(a, b; \psi).} \tag{4}$$

It holds on the complete product of angle circles, including literal zero coordinates. It permutes zero source weights exactly, keeps all state signs, and acts separately on source and testing coordinates. Real and imaginary state parts give the same conclusion for complex states. Thus $\pi \mathcal{C}_{\mathrm{can}} = \mathcal{C}_{\mathrm{can}}$. Since $c^\dagger = \pi \bar{c}$, (1) proves validity of (2). It remains to exclude its quartic completion.

2 An exact separating trace identity

For $\sigma = G, H$, let $N_{\sigma,0}, \dots, N_{\sigma,4}$ be the ten *integer* symmetric 4×4 matrices in Appendix A. Form the four integer block Hankel matrices

$$\mathsf{U}_\sigma = (N_{\sigma,i+j})_{i,j=0}^{2}, \qquad \mathsf{V}_\sigma = (N_{\sigma,i+j+1})_{i,j=0}^{1}. \tag{5}$$

They have orders $12, 8, 12, 8$ and are positive definite. Moreover,

$$(N_{G,k})_{\{1,2\},\{3,4\}} = (N_{H,k})_{\{1,2\},\{3,4\}} \qquad (0 \le k \le 4). \tag{6}$$

Positive definiteness follows from the rational pivots in Appendix A; the upper-right blocks agree entry by entry.

Write $Q(s) = \sum_{k=0}^{4} Q_k s^k$ for either completed matrix. Its half-line Gram representation is

$$Q(s) = Z_2(s)^T A_Q Z_2(s) + s Z_1(s)^T B_Q Z_1(s), \quad Z_2 = (I_4, sI_4, s^2 I_4)^T, \quad Z_1 = (I_4, sI_4)^T,$$

where $A_Q, B_Q \succeq 0$. Expanding gives

$$\sum_{k=0}^{4} \langle N_{\sigma,k}, Q_k \rangle = \langle \mathsf{U}_\sigma, A_Q \rangle + \langle \mathsf{V}_\sigma, B_Q \rangle. \tag{7}$$

Each J_k occurs in opposite off-diagonal blocks of G_k and H_k, so (6) cancels every selector coefficient. Consequently the sum of (7) is source-affine and nonnegative for every feasible certificate, including zero weights and singular Grams.

The resulting integer functional in the stated source order is

d	-97633735551856	λ	24154193783510329	n	85793705264
m	1787034292680750290	C	3303160925893294	A	16956553777510394
D	13682527570574	α	670221724556174	β	366431361010

(8)

Denote the functional with these coefficients by $\ell(c)$. Substituting P, N, E, K, with $v = 16s^2$, gives the exact identity

$$\boxed{12\ell(c) = \langle \mathsf{U}_G, A_G \rangle + \langle \mathsf{V}_G, B_G \rangle + \langle \mathsf{U}_H, A_H \rangle + \langle \mathsf{V}_H, B_H \rangle.} \tag{9}$$

Only the moments of degrees $0, 2, 4$ contribute to the nine source coefficients. If $Q_j^{(v)}(c, 0)$ is the coefficient of v^j with $J = 0$, the left side of (9) is

$$\sum_{j=0}^{2} 16^j \left[\langle N_{G,2j}, G_j^{(v)}(c,0) \rangle + \langle N_{H,2j}, H_j^{(v)}(c,0) \rangle \right]. \tag{10}$$

Thus $\ell \ge 0$ on $\mathcal{C}_{\mathrm{ram}}$. But direct rational evaluation yields

$$\boxed{\ell(c^\dagger) = -\frac{3260707096044735613817 2589}{9663676416000} < 0.} \tag{11}$$

The four PSD trace pairings in (9) cannot have a negative sum. This excludes the quartic selector.

To exclude every polynomial degree in the same coordinate, suppose that a polynomial matrix $J(s)$ made both completions PSD on $s \ge 0$. Their principal minors would give

$$|J_{ij}(s)|^2 \le (P + K)_{ii} (16s^2)(N + K)_{jj} (16s^2) = O(s^8).$$

The source diagonal blocks have degree at most four in s, so $\deg J_{ij} \le 4$. Every quartic matrix polynomial PSD on the half-line has the displayed Gram representation: factor the even globally PSD matrix polynomial $Q(t^2)$, average its polynomial SOS at t and $-t$, and separate the even and odd roots. With $s = t^2$, the even roots have the form $B_0 + sB_1 + s^2 B_2$ and the odd roots have the form $t(C_0 + sC_1)$, yielding $Z_2^T A_Q Z_2 + s Z_1^T B_Q Z_1$. Thus every polynomial selector would give the same four Gram blocks, already excluded by (11). This proves Theorem 1.

3 The contact mechanism and the remaining problem

The separation already holds at the exact contact source πc_* underlying (1). Its contact in positive angle gauges is

$$u = \frac{1}{16}, \qquad v = 13, \qquad \psi = \left(140, -\frac{1}{50}, \frac{1}{32}, 1\right),$$

and

$$\ell(\pi c_*) = -\frac{484433814365493288474548430934831 4410}{802811514300974151343419} < 0. \tag{12}$$

In the full state variables, write

$$\eta = L(1/16)\psi = \left(\frac{71681}{8192}, \frac{17}{6400}, \frac{279}{32}, -\frac{801}{800}\right).$$

At $U = 1/256$ and $s = \sqrt{13}/4$, any putative completion must satisfy

$$G(s)(\eta_1, \eta_2, U\eta_3, U\eta_4)^T = 0, \qquad H(s)\eta^T = 0. \tag{13}$$

These follow from the split identity and positivity of both summands at a zero. The separator excludes such a global quartic completion and remains negative after opening both zero event weights to the all-positive source (2).

The involution π preserves canonical validity and transports its testing functions between the parties. The five compact Bob weights in affine coordinates are

$$\frac{(1, \sqrt{|v|}/4, |v|/16, |v|^{1/4}/2, |v|^{3/4}/8)}{\sqrt{1+v^2}}.$$

Under (3), they become the fixed Alice weights

$$\boxed{\frac{(|u|, \sqrt{|u|}/4, 1/16, |u|^{3/4}/2, |u|^{1/4}/8)}{\sqrt{1+u^2}}}. \tag{14}$$

Their branches and normalization depend only on the test. The homogeneous pullback of the forty functions represents $b_0^2 F^h_{\pi c}$, where the original family represented $a_1^2 F^h_c$; both multipliers are nonnegative and positive on a dense testing chart.

The transported functions lie outside the original testing span. Restrict to $u > 0$, $v = 1$, $\psi = (1, 0, 0, 0)$. Every old compact feature is a constant times a polynomial of degree at most three divided by $(1+u^2)^{3/2}$, and is analytic at $u = 0$. One transported feature is

$$-\frac{u^{1/4}}{16\sqrt{2}\sqrt{1+u^2}}.$$

It has nonintegral order of vanishing and cannot lie in that span. Over the common multiplier $a_1^2 b_0^2$, multiply old roots by b_0 and transported roots by a_1; the same feature becomes $-u^{5/4}/(16\sqrt{2}(1+u^2))$, so the distinction persists.

In particular $\mathcal{C}_{\mathrm{ram}}$ and $\pi\mathcal{C}_{\mathrm{ram}}$ are incomparable: (1), (2), and $\pi^2 = 1$ give a point in each set difference. It remains open whether $\mathcal{C}_{\mathrm{ram}} + \pi\mathcal{C}_{\mathrm{ram}} = \mathcal{C}_{\mathrm{can}}$, or whether further testing functions are required.

A finite construction of the full cone requires a source-independent algebraic testing space whose Gram cone contains every valid slack, possibly after one fixed pure-testing multiplier. The contact and conductor identities and the qualitative tensor criterion of Ref. [3] do not specify such a global space.

References

[1] *Canonical Bell validity through two matrix inequalities*; *An exact obstruction to the quadratic half-line completion*, and the ramified order-49 data. Public repository, canonical-completion, commit 16b37f6.

[2] *Q222 Polar Assembly*, same commit, full-polar.

[3] Reconstructed R21 companion, same commit, latex, Chapters 3–6; in particular TF12, SG2, SG7, and CC1–CC6.

A The complete integer moment certificate

All matrices below are used as integers in (5)–(10). The normalized moment matrices are $N/(15 \cdot 2^{60})$; equations (5)–(10) use their integer numerators.

For each matrix R in (5), set

$$p_j = R_{jj} - \sum_{k<j} L_{jk}^2 p_k, \qquad L_{ij} = \frac{R_{ij} - \sum_{k<j} L_{ik} p_k L_{jk}}{p_j}, \quad L_{jj} = 1 \quad (i > j).$$

It gives $R = L\,\mathrm{diag}(p_j)L^T$. The exact floors of the smallest pivots are

R	U_G	V_G	U_H	V_H
$\lfloor \min_j p_j \rfloor$	794	12397	67408	1527738.

These positive pivot bounds prove that all four matrices are positive definite. The shared upper-right blocks in each pair $N_{G,k}, N_{H,k}$ make (6) directly visible.

$$N_{G,0} = \begin{pmatrix} 72283584904157745 & 98314305759870 & 214652490042795 & -146888491666410 \\ 98314305759870 & 135542238435 & 268778302935 & -201769656060 \\ 214652490042795 & 268778302935 & 1297549019460 & -401316089475 \\ -146888491666410 & -201769656060 & -401316089475 & 301979389605 \end{pmatrix}.$$

$$N_{G,1} = \begin{pmatrix} 29553765112790565 & 39441163448385 & 105424012102020 & -58918574211480 \\ 39441163448385 & 54072935055 & 110357620980 & -80768779620 \\ 105424012102020 & 110357620980 & 1079281903875 & -164509196205 \\ -58918574211480 & -80768779620 & -164509196205 & 120695415960 \end{pmatrix}.$$

$$N_{G,2} = \begin{pmatrix} 12789595044635020 & 15916490027610 & 66271058811705 & -23763057906660 \\ 15916490027610 & 21653142535 & 47712869220 & -32349910290 \\ 66271058811705 & 47712869220 & 1287764254705 & -70816288035 \\ -23763057906660 & -32349910290 & -70816288035 & 48333328645 \end{pmatrix}.$$

$$N_{G,3} = \begin{pmatrix} 6842765115107160 & 6582732635010 & 46453400242755 & -9802159230735 \\ 6582732635010 & 8693049165 & 22782075435 & -12984692925 \\ 46453400242755 & 22782075435 & 1981266595425 & -33639345030 \\ -9802159230735 & -12984692925 & -33639345030 & 19395668925 \end{pmatrix}.$$

$$N_{G,4} = \begin{pmatrix} 20910075570761184 & 4127281354275 & 33265473661770 & -6066895933245 \\ 4127281354275 & 3608024196 & 12294756105 & -5380775115 \\ 33265473661770 & 12294756105 & 5215970784456 & -19334942190 \\ -6066895933245 & -5380775115 & -19334942190 & 8026177596 \end{pmatrix}.$$

The five H moments

$$N_{H,0} = \begin{pmatrix} 51123677524583550 & -5656910689200 & 214652490042795 & -146888491666410 \\ -5656910689200 & 10186908915 & 268778302935 & -201769656060 \\ 214652490042795 & 268778302935 & 146089272033990 & -170629846948365 \\ -146888491666410 & -201769656060 & -170629846948365 & 2302846743948255 \end{pmatrix}.$$

$$N_{H,1} = \begin{pmatrix} 765887260600020 & 39305335635 & 105424012102020 & -58918574211480 \\ 39305335635 & 2703893715 & 110357620980 & -80768779620 \\ 105424012102020 & 110357620980 & 92030582539560 & -33880347103335 \\ -58918574211480 & -80768779620 & -33880347103335 & 160298762935095 \end{pmatrix}.$$

$$N_{H,2} = \begin{pmatrix} 89446398381610 & 50022371745 & 66271058811705 & -23763057906660 \\ 50022371745 & 931545700 & 47712869220 & -32349910290 \\ 66271058811705 & 47712869220 & 64202652830605 & -13709279518560 \\ -23763057906660 & -32349910290 & -13709279518560 & 18313581996550 \end{pmatrix}.$$

$$N_{H,3} = \begin{pmatrix} 48316110408750 & 24625867005 & 46453400242755 & -9802159230735 \\ 24625867005 & 328684920 & 22782075435 & -12984692925 \\ 46453400242755 & 22782075435 & 45819719138655 & -8466528163695 \\ -9802159230735 & -12984692925 & -8466528163695 & 3366601507110 \end{pmatrix}.$$

$$N_{H,4} = \begin{pmatrix} 33727791386796 & 13046747130 & 33265473661770 & -6066895933245 \\ 13046747130 & 116838834 & 12294756105 & -5380775115 \\ 33265473661770 & 12294756105 & 32917216546116 & -5858691614355 \\ -6066895933245 & -5380775115 & -5858691614355 & 1299098148456 \end{pmatrix}.$$